Unit III :: Conservation Laws

PHY01MX01R04: Mechanics and Properties of Matter

Dr. Parag Bhattacharya
Department of Physics
Rangapara College (Autonomous)

Syllabus

Unit III: Conservation Laws

Laboratory and centre-of-mass frames of reference; principle of conservation of linear momentum; collisions in one and two dimensions – elastic and inelastic; variable-mass systems, rocket propulsion; principle of conservation of angular momentum; applications: planetary motion, alpha-particle scattering, effect on linear and angular speeds of a particle upon orbital contraction, shape of galaxies.

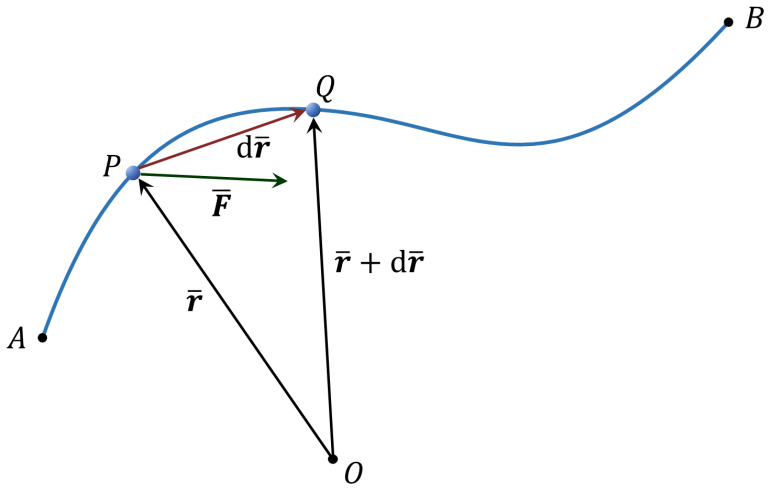
1 Concepts of work, energy and power

Before examining the work-energy theorem and conservation laws in mechanics, we must establish rigorous vector definitions for the foundational quantities of work, energy, and power:

- *Work:* an activity where a force acting on a system causes a displacement.
- *Energy:* the capacity of a system for doing work. A system may possess energy in the form of motion or position.
- *Kinetic energy:* the energy possessed by a system by virtue of its motion.
- *Potential energy:* the energy possessed by a system by virtue of its position or configuration.
- *Power:* the rate at which work is done or energy is used by a system.

1.1 Work

Let us consider a particle of mass m which is moving on an arbitrary path between points A and B. Suppose when the particle is at point P with a position vector $\mathbf{r}$, it is under the influence of some force $\mathbf{F}$ which causes it to get displaced to another point Q with a position vector $\mathbf{r}+\mathrm{d}\mathbf{r}$.



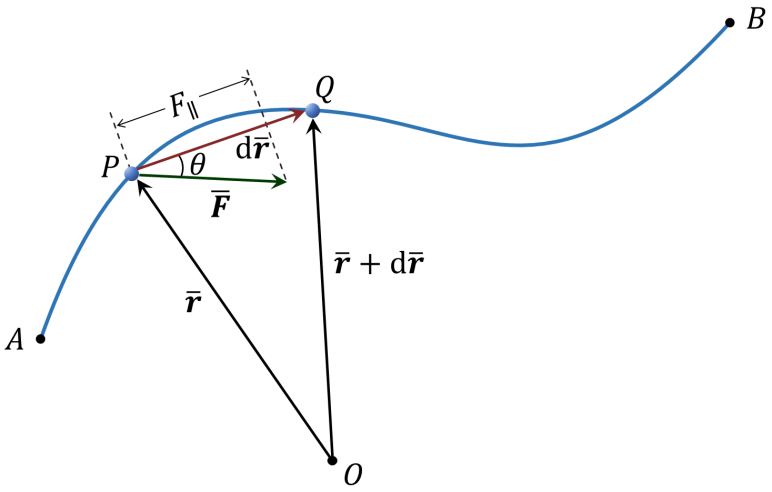
The infinitessimal amount of work done by the force $\mathbf{F}$ in moving the particle from P to Q is

$$\mathrm{d}W = \mathbf{F}\cdot\mathrm{d}\mathbf{r}$$

Hence, the total work done by the force $\mathbf{F}$ in moving the particle from A to B is

$$W = \int_A^B \mathrm{d}W = \int_A^B \mathbf{F}\cdot\mathrm{d}\mathbf{r}$$

1.1.1 Using projection of force vector



If the angle θ between **F** and d**r** is known, then we have

$$dW = F\cos\theta\, dr = F_\parallel\, dr$$

where $F_\parallel = F\cos\theta$ is the projection of **F** along d**r**. Then we can obtain the total work as

$$W = \int_A^B F\cos\theta\, dr = \int_A^B F_\parallel\, dr$$

1.1.2 Using components of force vector

We can write the force vector in component form as follows:

$$\mathbf{F} = \hat{\mathbf{x}}F_x + \hat{\mathbf{y}}F_y + \hat{\mathbf{z}}F_z$$

Also, the infinitessimal displacement is

$$d\mathbf{r} = \hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz$$

Then the infinitessimal displacement becomes

$$\begin{aligned} dW &= \mathbf{F}\cdot d\mathbf{r} \\ &= \left(\hat{\mathbf{x}}F_x + \hat{\mathbf{y}}F_y + \hat{\mathbf{z}}F_z\right)\cdot\left(\hat{\mathbf{x}}\,dx + \hat{\mathbf{y}}\,dy + \hat{\mathbf{z}}\,dz\right) \\ \implies dW &= F_x\,dx + F_y\,dy + F_z\,dz \end{aligned}$$

Therefore, the total work is obtained as

$$W = \int_A^B \left(F_x\,dx + F_y\,dy + F_z\,dz\right)$$

$$\implies W = \int_A^B F_x\,dx + \int_A^B F_y\,dy + \int_A^B F_z\,dz$$

1.1.3 Units of work

- In the SI system, the unit of work is the **joule** (or J), which is defined as the amount of work done by a force of 1 newton acting through a distance of 1 metre, i.e.

$$1\,\mathrm{J} = 1\,\mathrm{N} \times 1\,\mathrm{m}$$

- In the CGS system, the unit of work is the **erg**, which is defined as the amount of work done by a force of 1 dyne acting through a distance of 1 cm. In other words,

$$1\ \mathrm{erg} = 1\ \mathrm{dyne} \times 1\,\mathrm{cm}$$

  - The conversion to SI units are as follows:

$$1\ \mathrm{erg} = 10^{-7}\,\mathrm{J} \quad \text{and} \quad 1\,\mathrm{J} = 10^{7}\ \mathrm{erg}$$

- In the FPS system, the unit of work is the **foot-poundal** (or ft-lbf), which is defined as the amount of work done by a force of 1 poundal acting through a distance of 1 foot or

$$1\ \text{ft-lbf} = 1\ \text{lbf} \times 1\ \text{ft}$$

  - The conversion to SI units are as follows:

$$1\ \text{ft-lbf} \approx 1.356\,\mathrm{J} \quad \text{and} \quad 1\,\mathrm{J} \approx 0.736\ \text{ft-lbf}$$

1.2 Energy

In classical mechanics, energy is defined as the scalar capacity of a physical system to perform work. Rather than representing a tangible physical substance, energy is a fundamental, conserved state property that quantifies the state of a system relative to external interactions.

The units of energy are exactly the same as that of work.

1.2.1 Kinetic energy

Kinetic energy is the scalar energy possessed by an object due to its state of motion. For a particle of constant mass m moving at instantaneous velocity $\mathbf{v}$ relative to a reference frame, its kinetic energy is expressed as:

$$T = \frac{1}{2} m v^2$$

where $v = |\mathbf{v}| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$ is the instantaneous speed of the particle. Therefore, the kinetic energy of a particle is independent of the direction of motion of the particle.

1.2.2 Potential energy

Potential energy is stored energy which can be attibuted to the relative spatial configuration or position of a particle within a type of force field, known as a *conservative force*. In such a case, the conservative force may be expressed as the negative gradient of the potential energy:

$$\mathbf{F} = -\boldsymbol{\nabla} U$$

Common forms of potential energy include:

- Gravitational potential energy: $U_g = mgh$ (near Earth's surface)
- Elastic potential energy: $U_e = \frac{1}{2}kx^2$ (for an ideal spring satisfying Hooke's Law)

1.3 Power

Power is defined as the rate of doing work. When a force **F** displaces a particle by d**r** over a time interval $\mathrm{d}t$, the infinitessimal work done is $\mathrm{d}W = \mathbf{F}\cdot\mathrm{d}\mathbf{r}$. Then the instantaneous power delivered by the force is

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathbf{F}\cdot\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{F}\cdot\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$$
$$\implies P = \mathbf{F}\cdot\mathbf{v}$$

If some power P is delivered on a particle, the total work done on it is also obtained as

$$W = \int_{t_A}^{t_B} P\,\mathrm{d}t$$

where t_A, t_B are the instants of time when the particle was at positions A and B respectively.

1.3.1 Units of power

- In the SI system, the unit of power is the **watt** (or W), which is defined as the rate at which 1 J of work is done in 1 s, i.e.
$$1\,\mathrm{W} = 1\,\mathrm{J}/1\,\mathrm{s}$$
- In the CGS system, the unit of power is **erg per second** (or erg/s), which is defined as the rate at which 1 erg of work is done in 1 s.
  - The conversion to SI units are as follows:
$$1\text{ erg/s} = 10^{-7}\,\mathrm{W} \quad \text{and} \quad 1\,\mathrm{W} = 10^{7}\text{ erg/s}$$
- In the FPS system, the absolute unit of power is **foot-poundal per second** (or ft-lbf/s), which is defined as the rate at which 1 ft-lbf of work is done in 1 s.
  - The conversion to SI units are as follows:
$$1\text{ ft-lbf/s} \approx 1.356\,\mathrm{W} \quad \text{and} \quad 1\,\mathrm{W} \approx 0.736\text{ ft-lbf/s}$$
- In practical engineering applications, a commonly used unit of power from the FPS system is the **horsepower** (or hp), defined as
$$1\text{ hp} = 550\text{ ft-lbf/s}$$
  - The conversion to SI units are as follows:
$$1\text{ hp} \approx 0.746\,\mathrm{kW} \quad \text{and} \quad 1\,\mathrm{kW} \approx 1.341\text{ hp}$$

Exercises

1. Compute the work done in moving a particle from position $(0,1,2)$ to $(5,6,8)$ by the force $\mathbf{F} = \hat{\mathbf{x}}\left(2xy + z^2\right) + \hat{\mathbf{y}}x^2 + \hat{\mathbf{z}}2xz$.

Solution:

The given force is $\mathbf{F} = \hat{\mathbf{x}}\left(2xy + z^2\right) + \hat{\mathbf{y}}x^2 + \hat{\mathbf{z}}2xz$, from which we can identify that

$$F_x = 2xy + z^2,\ F_y = x^2,\ F_z = 2xz$$

Also, identifying the coordinates $(0,1,2)$ and $(5,6,8)$ to be those of the initial and final points A and B respectively, we have the total work done to be

$$\begin{aligned}
W &= \int_A^B F_x\,\mathrm{d}x + \int_A^B F_y\,\mathrm{d}y + \int_A^B F_z\,\mathrm{d}z \\
&= \int_A^B \left(2xy + z^2\right)\mathrm{d}x + \int_A^B x^2\,\mathrm{d}y + \int_A^B 2xz\,\mathrm{d}z \\
&= \int_A^B \left(2xy\,\mathrm{d}x + x^2\,\mathrm{d}y\right) + \int_A^B \left(z^2\,\mathrm{d}x + 2xz\,\mathrm{d}z\right) \\
&= \int_A^B \mathrm{d}\left(x^2y\right) + \int_A^B \mathrm{d}\left(xz^2\right) \\
&= x^2y\Big|_A^B + xz^2\Big|_A^B = \left[x^2y + xz^2\right]_A^B \\
&= x\left[xy + z^2\right]_{x=0,y=1,z=2}^{x=5,y=6,z=8} \\
&= \cdot 5 \cdot \left(5\cdot 6 + 8^2\right) - \cdot 0\left(0\cdot 1 + 2^2\right) \\
&= 5\cdot(30+64) - 0 \\
&= 5\times 94 \\
\therefore\ W &= 470\,\text{units}
\end{aligned}$$

2. A particle is moved from $(-1,0,2)$ to $(3,-8,4)$ by a force $\mathbf{F} = \hat{\mathbf{x}}\left(y^2 - x^2\right) + \hat{\mathbf{y}}2xy$. What is the total work done on the particle?

3. A particle lying in the xy-plane is acted upon by a force directed towards the origin whose magnitude is kr where $r = \sqrt{x^2 + y^2}$ is the distance from the particle to the origin.

   (a) Find the work that must be done to move the particle from the origin to the point $(1,1)$ along the radius vector to that point.

   (b) Find the work that must be done if the particle is first moved to $(1,0)$ and from there to $(1,1)$.

2 Work and energy

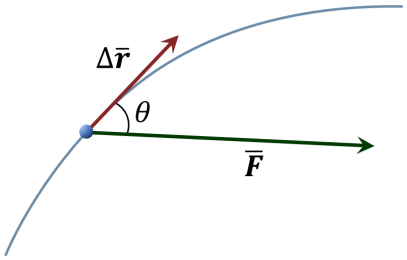
2.1 Work-energy theorem

As per Newton's second law, when a particle of mass m moving with a velocity $\mathbf{v}$ undergoes a force $\mathbf{F}$ at position $\mathbf{r}$, we have

$$\mathbf{F}(\mathbf{r}) = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$$

Suppose the particle undergoes a displacement of $\Delta\mathbf{r}$. Then the work done by the force is

$$\Delta W = \mathbf{F}\cdot\Delta\mathbf{r} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\Delta\mathbf{r} \tag{1}$$



For a time interval Δt which is small enough that $\mathbf{v}$ may be considered to be constant, we can write

$$\Delta\mathbf{r} = \mathbf{v}\Delta t$$

Then equation (1) becomes

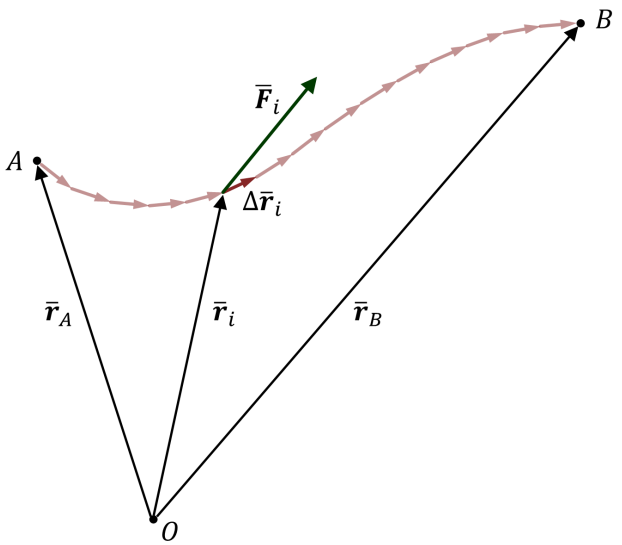
$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\Delta\mathbf{r} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v}\Delta t \tag{2}$$

Now, we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right) &= \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{v}\cdot\mathbf{v})\\ &= \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v} + \mathbf{v}\cdot\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = 2\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v}\\ \implies \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot\mathbf{v} &= \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right)\end{aligned} \tag{3}$$

Hence, from equations (1), (2) and (3), we have

$$\mathbf{F}\cdot\Delta\mathbf{r} = \frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right)\Delta t \tag{4}$$



Suppose now that the path from initial position A to final position B is divided into N small segments. Let $\mathbf{r}_A$ and $\mathbf{r}_B$ be the position vectors of the points A and B respectively.

Now, for the ith displacement at position $\mathbf{r}_i$, if the particle undergoes a force $\mathbf{F}_i$, then the work done over this displacement is obtained from equation (4) to be

$$\Delta W_i = \mathbf{F}_i\cdot\Delta\mathbf{r}_i = \frac{1}{2}m\frac{\mathrm{d}}{\mathrm{d}t}\left(v_i^2\right)\Delta t_i$$

Hence the total work done in moving the particle from A to B is obtained by summing over all the N displacements

$$W = \sum_{i=1}^{N} \Delta W_i = \sum_{i=1}^{N} \frac{1}{2} m \frac{\mathrm{d}}{\mathrm{d}t}\left(v_i^2\right) \Delta t_i \tag{5}$$

As the time intervals $\Delta t_i \to 0$, the summation in equation (5) becomes an integral and we have

$$\begin{aligned} W &= \int_{t_A}^{t_B} \frac{1}{2} m \frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right) \mathrm{d}t \\ &= \frac{1}{2} m \int_{v_A}^{v_B} \mathrm{d}\left(v^2\right) \\ \implies W &= \frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2 \end{aligned} \tag{6}$$

In equation (6), we recognise the terms on the RHS to be the kinetic energies of the particle at positions B and A respectively. Hence we can write

$$W = T_B - T_A$$

The above result may be generalised in the form of the *work-energy theorem*, which states that *the total work done on a system equals the change in its kinetic energy.*

$$\boxed{W_{\text{total}} = \Delta T}$$

3 Conservative and non-conservative forces

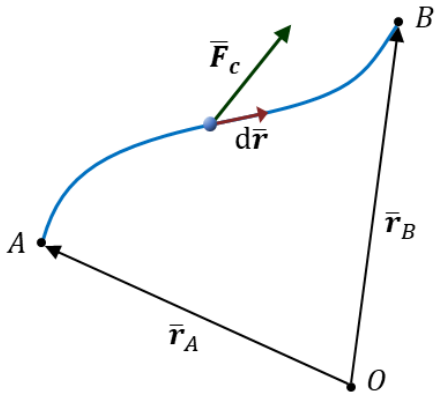
Based on the nature of the work done, we can classify different forces into two types:

1. *Conservative force:* A force is said to be conservative if the work done by it in displacing a particle between two arbitrary points depends only on the end points, and is independent of the actual path taken by the particle.
    - The net work done in a closed loop is always zero.
    - The mechanical energy (kinetic energy + potential energy) of the system is always conserved.
    - Examples of conservative forces:
        (a) Gravitational force – moving an object up and down takes or returns the exact same amount of energy, regardless of whether it is lifted straight up or taken by a winding ramp.
        (b) Elastic force – the force exerted by a stretched or compressed ideal spring depends solely on its displacement from the equilibrium position.
        (c) Electrostatic force – the force between two static electric charges depends only on their positions and magnitudes, not how they arrived there.
2. *Non-conservative force:* A force is said to be non-conservative if the work done by it in displacing a particle between two arbitrary points depends on the end points, as well as on the actual path taken by the particle.
    - The net work done in a closed loop is non-zero.
    - The mechanical energy (kinetic energy + potential energy) of the system is not conserved, but dissipated into other forms such as sound and heat.
    - Most non-conservative forces are *velocity-dependent*.
    - Examples of non-conservative forces:
        (a) Frictional force – the longer the path taken while pushing an object across a rough floor, the more energy is lost as heat.
        (b) Air resistance or drag force – the force opposing an object moving through the air, which increases with speed and surface area, while converting kinetic energy into thermal energy.
        (c) Viscous force or fluid resistance – the resistive force experienced by objects moving through liquids (such as honey or syrup).
        (d) Tension and compression forces – forces in ropes or cables that can dissipate energy through internal friction and material stretching.
        (e) Propulsion forces – forces exerted by engines, motors or muscles that actively consume chemical or electrical energy to produce motion.

3.1 Potential energy

Let us consider a particle being acted upon by a net conservative force $\mathbf{F}_c$ only. Then the total work done by $\mathbf{F}_c$ in moving an arbitrary particle from an initial point A (at position $\mathbf{r}_A$) to a final point B (at position $\mathbf{r}_B$) is

$$\int_{\mathbf{r}_A}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} = (\text{function of } \mathbf{r}_B) - (\text{function of } \mathbf{r}_A)$$



We refer to this function of position as the potential energy function U and write the above equation as follows

$$\int_{\mathbf{r}_A}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} = [-U(\mathbf{r}_B)] - [-U(\mathbf{r}_A)]$$

where the negative sign on U is to ensure the conservation of mechanical energy, which is observed experimentally.

Hence the above equation can be written as

$$\int_{\mathbf{r}_A}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} = U(\mathbf{r}_A) - U(\mathbf{r}_B) \tag{1}$$

We assume that this potential energy function vanishes at far-off distances from the source of the conservative force, i.e. $U \to 0$ as $\mathbf{r} \to \infty$.

Choosing $\mathbf{r}_A \to \infty$ and $\mathbf{r}_B = \mathbf{r}$, we obtain from equation (1)

$$\int_{\infty}^{\mathbf{r}} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} = \underbrace{U(\infty)}_{=0} - U(\mathbf{r})$$

$$\implies \boxed{U(\mathbf{r}) = -\int_{\infty}^{\mathbf{r}} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}} \tag{2}$$

Equation (2) shows that the potential energy of a particle at an arbitrary position equals the negative of the work done in moving that particle from a far-off distance to that position.

3.2 Conservative force as a negative gradient of potential energy

When a particle is acted upon by a net conservative force, the potential energy at an arbitrary point is a function of the position alone. So, for instance, in Cartesian coordinates we have $U(\mathbf{r}) = U(x, y, z)$. Hence the total differential of the potential energy is

$$\mathrm{d}U = \frac{\partial U}{\partial x}\mathrm{d}x + \frac{\partial U}{\partial y}\mathrm{d}y + \frac{\partial U}{\partial z}\mathrm{d}z$$

At far-off points, we have $U(\mathbf{r}\to\infty)=0$. Hence we can integrate the above from a far-off point to some arbitrary position $\mathbf{r}$ as follows

$$U(\mathbf{r}) = \int_0^{U(\mathbf{r})} \mathrm{d}U$$

$$\implies U(\mathbf{r}) = \int_\infty^{\mathbf{r}} \left[\frac{\partial U}{\partial x}\,\mathrm{d}x + \frac{\partial U}{\partial y}\,\mathrm{d}y + \frac{\partial U}{\partial z}\,\mathrm{d}z\right] \tag{1}$$

We can write the net conservative force in a Cartesian system as

$$\mathbf{F}_c = F_x\hat{\mathbf{x}} + F_y\hat{\mathbf{y}} + F_z\hat{\mathbf{z}} \tag{2}$$

Now, from the definition of the potential energy function, we have

$$U(\mathbf{r}) = -\int_\infty^{\mathbf{r}} \mathbf{F}_c\cdot\mathrm{d}\mathbf{r}$$

$$= -\int_\infty^{\mathbf{r}} \left(F_x\hat{\mathbf{x}} + F_y\hat{\mathbf{y}} + F_z\hat{\mathbf{z}}\right)\cdot\left(\hat{\mathbf{x}}\,\mathrm{d}x + \hat{\mathbf{y}}\,\mathrm{d}y + \hat{\mathbf{z}}\,\mathrm{d}z\right)$$

$$\implies U(\mathbf{r}) = -\int_\infty^{\mathbf{r}} \left(F_x\,\mathrm{d}x + F_y\,\mathrm{d}y + F_z\,\mathrm{d}z\right) \tag{3}$$

Comparing equations (1) and (3), we get

$$F_x = -\frac{\partial U}{\partial x},\quad F_y = -\frac{\partial U}{\partial y},\quad F_z = -\frac{\partial U}{\partial z}$$

Hence from equation (2), we obtain

$$\mathbf{F}_c = -\frac{\partial U}{\partial x}\hat{\mathbf{x}} - \frac{\partial U}{\partial y}\hat{\mathbf{y}} - \frac{\partial U}{\partial z}\hat{\mathbf{z}}$$

$$\implies \mathbf{F}_c = -\left[\hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}\right]U \tag{4}$$

The *nabla operator* $\boldsymbol{\nabla}$ is a vector differential operator which is defined as

$$\boldsymbol{\nabla} = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}$$

When $\boldsymbol{\nabla}$ acts on some scalar field $f(\mathbf{r})$, we obtain the *gradient* of the field $\boldsymbol{\nabla} f$.

Hence, equation (4) can be written as

$$\boxed{\mathbf{F}_c(\mathbf{r}) = -\boldsymbol{\nabla}U(\mathbf{r})}$$

Therefore, a conservative force can be expressed as the negative gradient of the potential energy.

3.3 Law of conservation of mechanical energy

Consider an arbitrary particle which is moving from an initial point A to a final point B (with position vectors $\mathbf{r}_A$ and $\mathbf{r}_B$ respectively) under the influence of conservative forces only. Suppose the net conservative force is $\mathbf{F}_c$.

We know that the potential energy at an arbitrary point $\mathbf{r}$ is given as

$$U(\mathbf{r}) = -\int_{\infty}^{\mathbf{r}} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}$$

Then the potential energies at the positions of A and B are

$$U_A = -\int_{\infty}^{\mathbf{r}_A} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} \quad \text{and} \quad U_B = -\int_{\infty}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}$$

Hence the change in the potential energy during the entire motion of the particle is

$$\begin{aligned}
\Delta U &= U_B - U_A \\
&= \left[-\int_{\infty}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}\right] - \left[-\int_{\infty}^{\mathbf{r}_A} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}\right] \\
&= \left[-\int_{\infty}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}\right] - \left[\int_{\mathbf{r}_A}^{\infty} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}\right] \\
&= -\int_{\infty}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} - \int_{\mathbf{r}_A}^{\infty} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} \\
&= -\left[\int_{\mathbf{r}_A}^{\infty} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} + \int_{\infty}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r}\right] \\
\implies \Delta U &= -\int_{\mathbf{r}_A}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} \qquad (1)
\end{aligned}$$

The total work done by the conservative forces in moving the particle from A to B is

$$W = \int_{\mathbf{r}_A}^{\mathbf{r}_B} \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} \qquad (2)$$

$$\text{also,} \quad W = -\Delta U \quad [\text{using eqn. (1)}]$$

From the work-energy theorem, we know that the total work done equals the change in kinetic energy of the particle, i.e.

$$W = \Delta T = T_B - T_A \qquad (3)$$

where T_A and T_B are the kinetic energies of the particle at points A and B respectively.

Hence, from equations (1), (2) and (3), we obtain

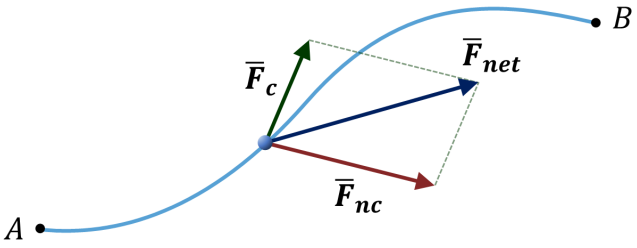
$$\begin{aligned} \Delta U &= -\Delta T \\ \implies \Delta T + \Delta U &= 0 \\ \implies (T_B - T_A) + (U_B - U_A) &= 0 \\ \implies T_A + U_A &= T_B + U_B \\ \implies E_A &= E_B \\ \implies \boxed{E = \text{constant}} & \end{aligned}$$

Therefore, the mechanical energy (which is the sum of the kinetic and potential energies) remains constant when a particle moves under the influence of conservative forces alone.

This is known as the *law of conservation of mechanical energy*.

3.4 Work done by non-conservative forces

Consider a particle, moving along an arbitrary path from point A to point B, as shown below.



Suppose the resultant of all conservative forces is $\mathbf{F}_c$ and the resultant of all non-conservative forces is $\mathbf{F}_{nc}$. Then the net force acting on the particle at any point on its path is

$$\mathbf{F}_{\text{net}} = \mathbf{F}_c + \mathbf{F}_{nc} \tag{1}$$

Then the total work done by all the forces in moving the particle along the path is

$$\begin{aligned} W_{\text{total}} &= \int_A^B \mathbf{F}_{\text{net}} \cdot \mathrm{d}\mathbf{r} \\ &= \int_A^B (\mathbf{F}_c + \mathbf{F}_{nc}) \cdot \mathrm{d}\mathbf{r} \quad \text{[using eqn. (1)]} \\ &= \int_A^B \mathbf{F}_c \cdot \mathrm{d}\mathbf{r} + \int_A^B \mathbf{F}_{nc} \cdot \mathrm{d}\mathbf{r} \\ \implies W_{\text{total}} &= W_c + W_{nc} \end{aligned} \tag{2}$$

In equation (2), W_c and W_{nc} are the work done separately on the particle by the conservative and the non-conservative forces respectively.

From the work-energy theorem, the change in the kinetic energy of the particle equals the total work done on it, i.e.

$$W_{\text{total}} = \Delta T \tag{3}$$

Also, the change in the potential energy of the particle equals the negative of the work done on it by the conservative forces, i.e.

$$W_c = -\Delta U \tag{4}$$

Hence, from equations (2), (3) and (4), we obtain

$$\begin{aligned} \Delta T &= -\Delta U + W_{nc} \\ \implies W_{nc} &= \Delta T + \Delta U \\ &= \Delta(T+U) \\ \implies &\boxed{W_{nc} = \Delta E} \end{aligned} \tag{5}$$

Therefore, the work done on a system by all the non-conservative forces equals the change in the mechanical energy of the system.

This is known as the *general law of conservation of energy*.

Special case

Suppose we have a system in which non-conservative forces are either absent or do no work. In either case, we would have $W_{nc} = 0$. In such a system, only the conservative forces do any work.

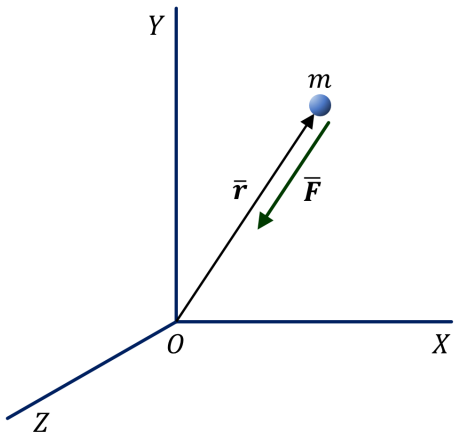
Then from equation (5), we obtain

$$\begin{aligned} \Delta E &= 0 \\ \implies E &= \text{constant} \end{aligned}$$

Therefore, this special case reduces to the law of conservation of mechanical energy.

4 Linear restoring force

A linear restoring force is one whose magnitude is linearly proportional to the displacement of a particle from a fixed reference point that is its *mean position* of rest or equilibrium, and direction is opposite to the displacement.



If a particle of mass m, which is at a point with position vector $\mathbf{r}$, is under the influence of a linear restoring force such that its mean position is the origin, then we have

$$\begin{aligned} &\mathbf{F} \propto (-\mathbf{r}) \\ \text{or,}\quad &\mathbf{F} = -k\mathbf{r} \end{aligned} \tag{1}$$

where k is the proportionality constant, known as the *force constant*.

The linear restoring force is a conservative force and hence it has an associated potential energy field which can be obtained at an arbitrary point $\mathbf{r}$ as

$$U(\mathbf{r}) = -\int_{\infty}^{\mathbf{r}} \mathbf{F}\cdot \mathrm{d}\mathbf{r} \tag{2}$$

From equation (1), we can write

$$\mathbf{F} = -k\left(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}\right)$$

Also, we have that

$$\mathrm{d}\mathbf{r} = \hat{\mathbf{x}}\,\mathrm{d}x + \hat{\mathbf{y}}\,\mathrm{d}y + \hat{\mathbf{z}}\,\mathrm{d}z$$

This gives us

$$\begin{aligned} &\mathbf{F}\cdot\mathrm{d}\mathbf{r} = -k\left(x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}\right)\cdot\left(\hat{\mathbf{x}}\,\mathrm{d}x + \hat{\mathbf{y}}\,\mathrm{d}y + \hat{\mathbf{z}}\,\mathrm{d}z\right) \\ \implies &\mathbf{F}\cdot\mathrm{d}\mathbf{r} = -k\left(x\,\mathrm{d}x + y\,\mathrm{d}y + z\,\mathrm{d}z\right) \end{aligned}$$

Hence, in equation (2), we have

$$\begin{aligned} U(\mathbf{r}) &= -\int_{\infty}^{\mathbf{r}} -k\left(x\,\mathrm{d}x + y\,\mathrm{d}y + z\,\mathrm{d}z\right) \\ &= k\left[\int_{\infty}^{x} x\,\mathrm{d}x + \int_{\infty}^{y} y\,\mathrm{d}y + \int_{\infty}^{z} z\,\mathrm{d}z\right] \\ &= k\left(\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2}\right) + C \\ &= \frac{1}{2}k\left(x^2 + y^2 + z^2\right) + C \\ \implies U(\mathbf{r}) &= \frac{1}{2}kr^2 + C \end{aligned} \tag{3}$$

where $C = k\left[\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2}\right]_{x,y,z\to\infty}$.

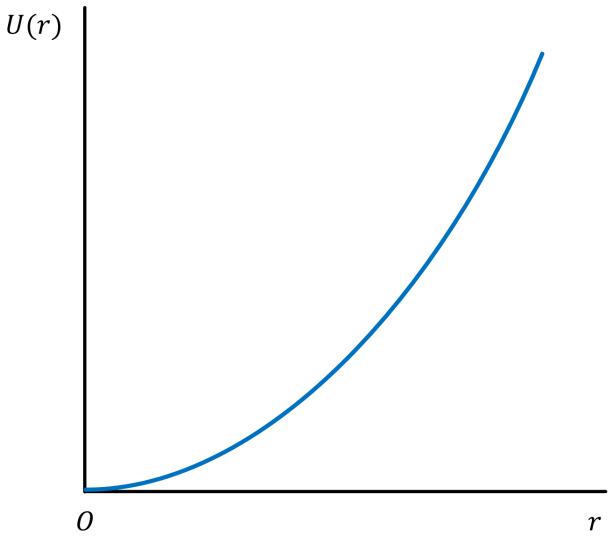
If we choose the potential energy to be zero at the origin, i.e. $U = 0$ at $\mathbf{r} = \mathbf{0}$, then from equation (3), we obtain

$$0 = \frac{1}{2}k \times 0 + C$$
$$\implies C = 0$$

Then equation (3) becomes

$$\boxed{U(\mathbf{r}) = \frac{1}{2}kr^2} \qquad (4)$$

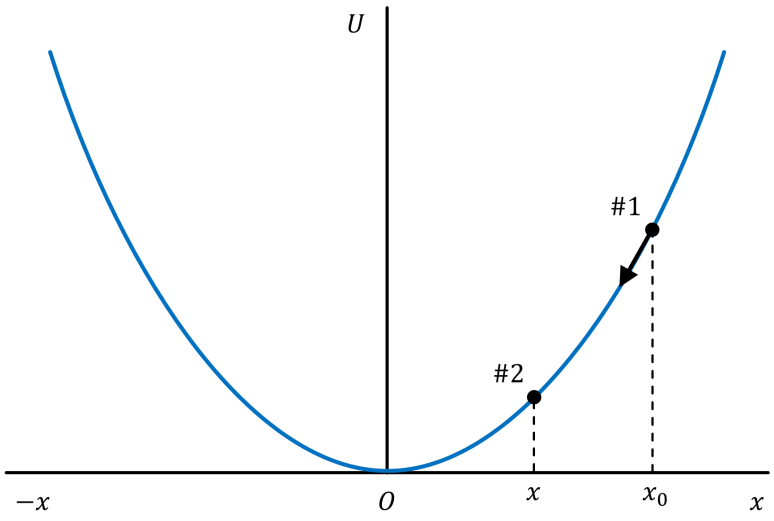
Equation (4) shows that the potential energy due to a linear restoring force is a quadratic function of the distance of the particle from the mean position.



Suppose the motion of the particle is confined to the X-axis alone, i.e. $\mathbf{r} = x\hat{\mathbf{x}}$ and hence $r^2 = x^2$. Then the restoring force is

$$\mathbf{F} = -kx\hat{\mathbf{x}}$$

The potential energy at any point on the X-axis is $U = \frac{1}{2}kx^2$.



Let the particle be initially at the extreme position x_0 from the mean position at the origin, which corresponds to #1 on the $U(x)$ plot. Then the initial potential energy at #1 is

$$U_1 = \frac{1}{2}kx_0^2$$

If the particle is released from position #1 from rest, its initial kinetic energy is

$$T_1 = 0$$

Suppose that finally the particle reaches some arbitrary position x, which corresponds to #2 on the $U(x)$ plot. Then the final potential energy is

$$U_2 = \frac{1}{2}kx^2$$

Because at position #2, the particle is in motion, its final kinetic energy is

$$T_2 = \frac{1}{2}mv^2$$

Hence, the changes in its kinetic and potential energies are

$$\Delta T = T_2 - T_1 = \frac{1}{2}mv^2 - 0 = \frac{1}{2}mv^2$$
$$\Delta U = U_2 - U_1 = \frac{1}{2}kx^2 - \frac{1}{2}kx_0^2 = \frac{1}{2}k\left(x^2 - x_0^2\right)$$

Because the linear restoring force is conservative, we obtain from the law of conservation of mechanical energy that

$$\Delta T + \Delta U = 0$$
$$\implies \frac{1}{2}mv^2 + \frac{1}{2}k\left(x^2 - x_0^2\right) = 0$$
$$\implies v^2 = \frac{k}{m}\left(x_0^2 - x^2\right)$$
$$\implies \boxed{v = \pm\sqrt{\frac{k}{m}\left(x_0^2 - x^2\right)}} \tag{5}$$

From Newton's second law of motion, we obtain the acceleration as

$$F = ma = -kx$$
$$\implies \boxed{a = -\frac{k}{m}x} \tag{6}$$

- When the particle is at the extreme positions at either side, i.e. $x = \pm x_0 \implies x^2 = x_0^2$, then we obtain from equation (5) that

$$v = \pm\sqrt{\frac{k}{m}\left(x_0^2 - x_0^2\right)}$$
$$\implies v = 0$$

  Also, from equation (6), we obtain

$$a = a_{\max} = -\frac{k}{m}x_0$$

  This means that the particle is at rest but under maximum magnitude of acceleration when it is at either of the extreme positions.

- When the particle is at the mean position, i.e. $x = 0$, then we obtain from equation (5) that

$$v = \pm\sqrt{\frac{k}{m}\left(x_0^2 - 0\right)}$$

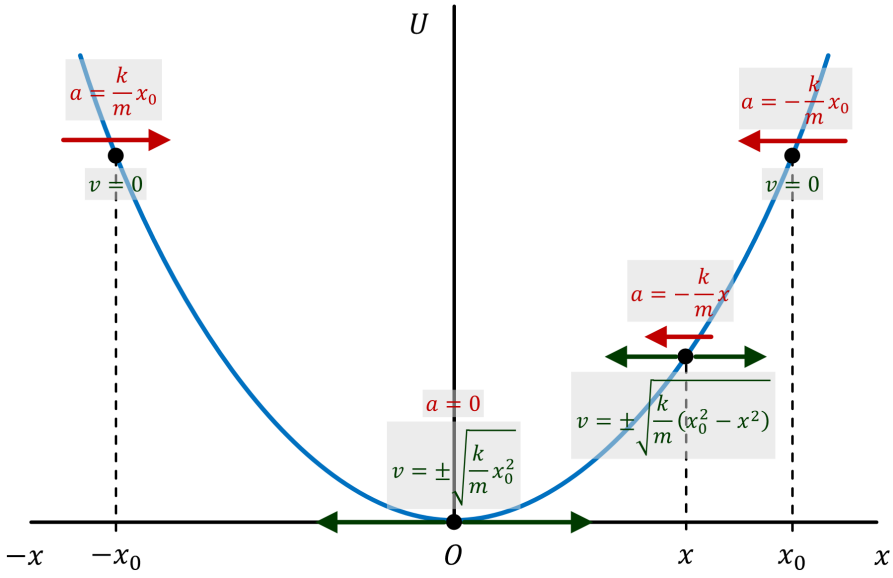
$$\implies v = v_{\max} = \pm\sqrt{\frac{k}{m}x_0^2}$$

And from equation (6), we obtain

$$a = 0$$

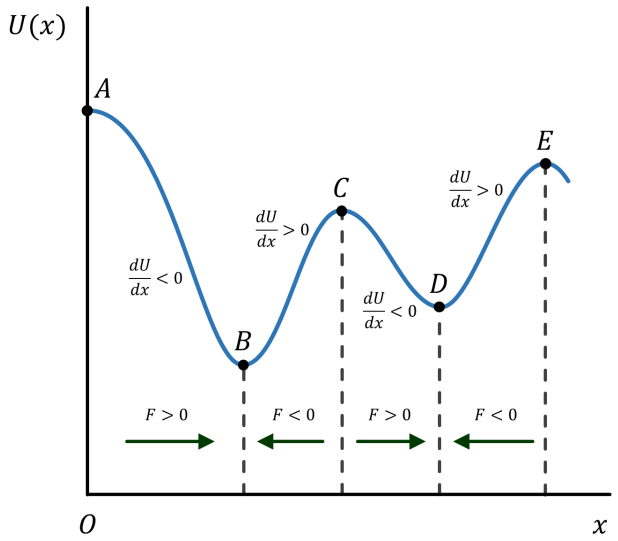
This means that the particle moves with a maximum speed but under zero acceleration when it is at the mean positions.

The acceleration and velocities of the particle at the extreme positions, mean position and an arbitrary position are shown in the figure below.



5 Potential energy and equilibria

In a conservative force field, the potential energy U of a particle is a function of its position in space. A plot showing the variation of potential energy U of a particle with its position (e.g. along the x-axis) is known as a *potential energy curve*. This curve provides crucial information about the motion, forces and stability of the particle without requiring the complete solution of its equations of motion.



For one-dimensional motion along the x-axis, a conservative force F is obtained from the potential energy function $U(x)$ by the relation:

$$F = -\frac{\mathrm{d}U}{\mathrm{d}x}$$

Thus, the force acting on a particle at any point is equal to the negative slope of the potential energy curve at that point.

- *Positive slope* ($\mathrm{d}U/\mathrm{d}x > 0$): In regions, such as BC and DE, where U increases as x increases, the slope $\dfrac{\mathrm{d}U}{\mathrm{d}x}$ is positive. Consequently, the force F is negative, i.e. directed towards the left or in the direction of decreasing x.
- *Negative slope* ($\mathrm{d}U/\mathrm{d}x < 0$): In regions, such as AB and CD, where U decreases as x increases, the slope $\dfrac{\mathrm{d}U}{\mathrm{d}x}$ is positive. Consequently, the force F is positive, i.e. directed towards the right or in the direction of increasing x.

In either case, the force acting on the particle always tends to move it towards a region of lower potential energy.

5.1 Concept of equilibrium

A particle is said to be in a position of equilibrium when the net force acting on it is zero, i.e.

$$\mathbf{F}_{\text{net}} = \mathbf{0}$$

Let us consider a conservative force F acting along the x-axis. Then we have

$$F = -\frac{\mathrm{d}U}{\mathrm{d}x}$$

If a particle under the influence of the force F is in equilibrium at some point x_0, then we have

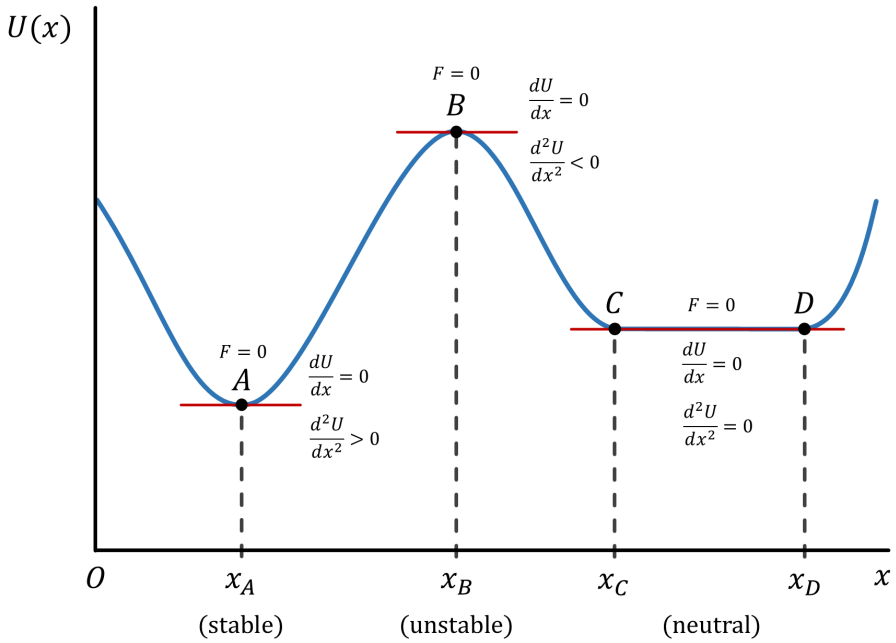
$$F(x_0) = 0$$

$$\implies \left.\frac{\mathrm{d}U}{\mathrm{d}x}\right|_{x=x_0} = 0$$

On a potential energy curve $U(x)$, the slope at the point of equilibrium x_0 is zero. Hence, it could either be either on a *local maxima* or on a *local minima* or on a *flat horizontal region.*

5.2 Types of equilibria

The points of equilibria are broadly classified into three types based on how the system responds to a small displacement from the equilibrium position.



1. **Stable equilibrium:**

   - If a particle, which is at a point of stable equilibrium, is slightly displaced in either direction, the resulting force acts in a direction which is opposite to the displacement.
   - The force acts as a restoring force which pulls the particle back towards its equilibrium position.
   - Points of stable equilibria occur at local minima, such as point A in the figure above, i.e. $x = x_A$.
   - At a point of stable equilibrium, we have

$$\frac{\mathrm{d}U}{\mathrm{d}x} = 0 \quad \text{and} \quad \frac{\mathrm{d}^2U}{\mathrm{d}x^2} > 0$$

2. **Unstable equilibrium:**

   - If a particle, which is at a point of unstable equilibrium, is slightly displaced in either direction, the resulting force acts in a direction which is along the displacement.
   - The force pushes the particle further away from its equilibrium position.
   - Points of unstable equilibria occur at local maxima, such as point B in the figure above, i.e. $x = x_B$.

- At a point of unstable equilibrium, we have

$$\frac{\mathrm{d}U}{\mathrm{d}x}=0 \quad \text{and} \quad \frac{\mathrm{d}^2U}{\mathrm{d}x^2}<0$$

3. **Neutral equilibrium:**

- If a particle, which is at a point of neutral equilibrium, is slightly displaced in either direction, it remains at rest in its new position without tending either to return to its initial position or to move further away.
- The force remains zero over the entire region of neutral equilbrium..
- Points of neutral equilibria occur over flat horizontal regions, such as the region between points C and D in the figure above, i.e. $x_C < x < x_D$.
- At a point of neutral equilibrium, we have

$$\frac{\mathrm{d}U}{\mathrm{d}x}=0 \quad \text{and} \quad \frac{\mathrm{d}^2U}{\mathrm{d}x^2}=0$$